

MA Θ Competition Team Problem Set 17

Anders Christensen, Minjoo Kim

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Problem 1. Consider the following premises:

- I. All Danes have surnames ending with "-sen"
- II. Some people with the surname Christensen are Danish

Can you conclude that all people with the surname Christensen are Danish? Why or why not?

Problem 2. All poets are dreamers. Some dreamers are not logical. All logical people are mathematicians. Which of the following must be true?

- I. Some poets are mathematicians
- II. Some poets are not logical
- III. Some dreamers are mathematicians
- IV. No mathematicians are poets
- V. None must be true

Problem 3. On a certain unnamed island, each person is either a truth-teller (always tells the truth) or a liar (always lies).

Jeff says "Bill is a liar." Bill says "Jeff and I are of opposite types."
What are Jeff and Bill?

- I. Both truth-tellers
- II. Both liars
- III. Jeff truth-teller, Bill liar
- IV. Jeff liar, Bill truth-teller
- V. Cannot be determined

Problem 4. Consider the following series of statements:

- I. Exactly one of these statements is true.
- II. Exactly two of these statements are true.
- III. Exactly three of these statements are true.

How many are true?

Problem 5. The Mu Alpha Theta execs have had enough of members trying to sneak out during meetings, so they booby-trapped one of the two doors in Mr. Warkentin's room. Dom and Anders know which is which. Dom will always tell the truth, while Anders will always lie. You can only ask one question to one exec. How do you find the safe door and escape Mu Alpha Theta?

Problem 6. If all alligators are ferocious creatures and some creepy crawlers are alligators, which statement(s) must be true?

- I. All alligators are creepy crawlers.
 - II. Some ferocious creatures are creepy crawlers.
 - III. Some alligators are not creepy crawlers.
- (A) I only (B) II only (C) III only (D) II and III only (E) None must be true

Problem 7. Ami's Airport Taxi brought five passengers to the airport yesterday, each leaving on a different flight.

Using the clues below, determine the passenger name, destination, and purpose for travel for each of the departure times in the table on the next page.

- I. The five people are Helenka, Iveta, the person visiting family, the person going to Tokyo, and the person leaving at 5:15 p.m.
- II. Tuur left one hour after the person going to Rome, who was travelling for school.
- III. Mijo left at 1:15 p.m. and was going on vacation. He was not the person going to Halifax.
- IV. Gaurav and Helenka were two of the passengers. Only one of them went to Vancouver.
- V. Two people were traveling for business. They left five hours apart and went to Vancouver and Paris, in some order.
- VI. Fill in the table with the passenger name, destination, and purpose for travel for each of the given departure times.

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